BINOMIAL THEOREM

$$(x+y)^n = {}^nC_0x^ny^0 + {}^nC_1x^{n-1}y^1 + \cdots + {}^nC_nx^0y^n$$

$$= \sum_{r=0}^{n} {}^nC_r\, x^{n-r}y^r$$

no. of terms : $n+1$

☆ General term : $T_{r+1} = {}^nC_r\, x^{n-r}y^r$

☆ p^{th} term from end $= (n-p+2)^{th}$ from beginning

Pascal Δ

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1

Middle term

(i) n odd → $\left(\frac{n+1}{2}\right)^{th}$ and $\left(\frac{n+3}{2}\right)^{th}$ term

(ii) n even → $\left(\frac{n}{2}+1\right)^{th}$ term

☆ $(ax+by)^n$: $T_{r+1} = \boxed{{}^nC_r\, a^{n-r}b^r}\, x^{n-r}y^r$ coeff = pehla hissa

☆ Binomial coeff of middle term is always greatest.

(iii) n even → greatest coeff ${}^nC_{n/2}$; n odd → ${}^nC_{\frac{n-1}{2}} = {}^nC_{\frac{n+1}{2}}$

Ex. $(1+x)^{2n}$ me greatest coeff x^n ka ${}^{2n}C_n$; $(1+x)^9$: ${}^9C_4 =$ (126)

Numerically greatest term : $(a+b)^n$

Let $m = \dfrac{n+1}{1+\left|\frac{a}{b}\right|}$

(i) m integer → $T_m = T_{m+1}$ (dono greatest)

(ii) m not integer → $T_{[m]+1}$ greatest

Ex. $(3+2x)^9$ at $x=1$: $m = \dfrac{10}{1+3/2} = 4 \Rightarrow T_4 = T_5 =$ (489888)

Ex. $(1+x)^{10}$ at $x=\frac{1}{2}$: $m = \dfrac{11}{1+2} = 3.67 \Rightarrow T_4 = {}^{10}C_3/2^3 =$ (15)

Properties of Binomial coeff.

(i) ${}^nC_r = {}^nC_{n-r}$

(ii) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

(iii) ${}^nC_n + {}^{n+1}C_n + \cdots + {}^{n+k}C_n = {}^{n+k+1}C_{n+1}$

☆☆ (iv) $r \cdot {}^nC_r = n \cdot {}^{n-1}C_{r-1}$

☆☆ (v) $(r+1) \cdot {}^{n+1}C_{r+1} = (n+1) \cdot {}^nC_r$ inse "r" hata do

(vi) $\frac{{}^nC_{r+1}}{{}^nC_r} = \frac{n-r}{r+1}$; $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$

(vii) ${}^nC_x = {}^nC_y \Rightarrow x = y$ or $x + y = n$

Imp. Series Expansions

☆ (i) $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$

(ii) $(x+1)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} + \cdots + {}^nC_n$

(iii) $(1-x)^n = {}^nC_0 - {}^nC_1 x + {}^nC_2 x^2 - \cdots + (-1)^n {}^nC_n x^n$

(iv) $(x-1)^n = {}^nC_0 x^n - {}^nC_1 x^{n-1} + \cdots + (-1)^n {}^nC_n$

Examples on expansion

Ex. $(x-2)^5 = x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$

Ex. ${}^nC_8 = {}^nC_{12} \Rightarrow n = 8 + 12 = \boxed{20}$

Ex. Coeff of x^3 in $(1+x)^{10} = {}^{10}C_3 = \boxed{120}$

Ex. Coeff of x^r in $(1+x)^n$ = ways of choosing r out of n

Identity prove karni ho to dono taraf same cheez do tareeke se ginte hain

Examples on terms

Ex. Middle term of $\left(x + \frac{1}{x}\right)^8 = T_5 = {}^8C_4 = \boxed{70}$

Ex. Greatest binomial coeff in $(1+x)^9$: ${}^9C_4 = {}^9C_5 = \boxed{126}$

Ex. $(x+a)^{10}$ me end se 4th term = shuru se $10 - 4 + 2 = \boxed{8^{th}}$

Ex. $(x+a)^{100} + (x-a)^{100}$ me total terms = $\boxed{51}$

$(x+a)^n \pm (x-a)^n$: n even $\rightarrow \frac{n}{2} + 1$, n odd $\rightarrow \frac{n+1}{2}$ terms

Ratio & rational terms

(i) $\dfrac{T_{r+1}}{T_r} = \dfrac{n-r+1}{r} \cdot \dfrac{a}{b}$ for $(a+b)^n$

(ii) $(a^{1/p} + b^{1/q})^n$: $T_{r+1} = {}^nC_r\, a^{\frac{n-r}{p}} b^{\frac{r}{q}}$ $\Rightarrow$ $\frac{n-r}{p}, \frac{r}{q}$ both integers

(iii) $(x+a)^n - (x-a)^n$: n even $\to$ $\frac{n}{2}$, n odd $\to$ $\frac{n+1}{2}$ terms

Ex. $(\sqrt{2} + \sqrt[3]{3})^{100}$: r even and $3 \mid r \Rightarrow r = 0, 6, \ldots, 96$ $\Rightarrow$ (17 terms)

Consecutive coefficients

(i) Coeffs of T_r, T_{r+1}, T_{r+2} in A.P. $\iff n^2 - (4r+1)n + 4r^2 - 2 = 0$

(ii) Ratio $1:2:3$: $\frac{n-r+1}{r} = 2$, $\frac{n-r}{r+1} = \frac{3}{2} \Rightarrow n = 14$

Ex. $n = 7$: 7, 21, 35 are in A.P. (difference 14)

Ex. ${}^{14}C_4 : {}^{14}C_5 : {}^{14}C_6 = 1001 : 2002 : 3003 =$ (1 : 2 : 3)

Finding a term : T_{r+1} likho, power barabar karo

Ex. Independent of x in $\left(x^2 + \dfrac{1}{x}\right)^9$: $T_{r+1} = {}^9C_r\, x^{18-3r}$, $r = 6 \Rightarrow$ (84)

Ex. Coeff of x^5 in $\left(x^2 + \dfrac{2}{x}\right)^7$: $T_{r+1} = {}^7C_r 2^r x^{14-3r}$, $r = 3 \Rightarrow$ (280)

Ex. Rational terms in $(\sqrt{2} + \sqrt[3]{3})^{100}$: r multiple of 6 $\Rightarrow$ (17)

r whole number, $0 \le r \le n$ hona chahiye (warna term nahi hai)

Sum of coefficients : In (i)

(a) Put $x = 1$: $C_0 + C_1 + C_2 + \cdots + C_n = 2^n$

(b) Put $x = -1$: $C_0 - C_1 + C_2 - \cdots + (-1)^n C_n = 0$

(c) (a)+(b) $\Rightarrow$ $C_0 + C_2 + C_4 + \cdots = 2^{n-1}$

(d) (a)−(b) $\Rightarrow$ $C_1 + C_3 + C_5 + \cdots = 2^{n-1}$

(e) Put $x = i$ and $x = -i$, add : $C_0 - C_2 + C_4 - \cdots = \dfrac{(1+i)^n + (1-i)^n}{2}$

(f) Subtract : $C_1 - C_3 + C_5 - \cdots = \dfrac{(1+i)^n - (1-i)^n}{2i}$

Ex. $\displaystyle\sum_{r=0}^{6} (-1)^r\, {}^{13}C_r = {}^{12}C_6 =$ (924) half sum 0 nahi hota !

Coefficient of x^k in a sum of powers

(i) $(1+x)^p + \cdots + (1+x)^q = \dfrac{(1+x)^{q+1} - (1+x)^p}{x}$

(ii) coeff of $x^k = {}^{q+1}C_{k+1} - {}^{p}C_{k+1}$

Ex. x^3 in $(1+x)^5 + \cdots + (1+x)^9 = {}^{10}C_4 - {}^{5}C_4 =$ (205)

(iii) Hockey stick : $\sum_{k=r}^{n} {}^{k}C_r = {}^{n+1}C_{r+1}$

Ex. $\sum_{k=2}^{9} {}^{k}C_2 = {}^{10}C_3 =$ (120)

Ex. Constant term of $(1+x)^4\left(1+\dfrac{1}{x}\right)^4 = \sum({}^{4}C_r)^2 = {}^{8}C_4 =$ (70)

Sum of coefficients of a polynomial

(i) Sum of all coeffs of $f(x)$ = $f(1)$

(ii) Even-power coeffs sum $= \dfrac{f(1)+f(-1)}{2}$; odd-power $= \dfrac{f(1)-f(-1)}{2}$

Ex. $(1+x+x^2)^5$: $f(1) = 243,\ f(-1) = 1 \Rightarrow$ even-power sum $=$ (122)

Ex. $(2x-3y)^6$ ke saare coeffs ka sum $= (2-3)^6 =$ (1)

(iii) Differentiate $(1+x)^n$: $n(1+x)^{n-1} = \sum r\,{}^{n}C_r\,x^{r-1}$

(iv) Put $x = 2$: $\sum 2^r\,{}^{n}C_r = 3^n$, $x = 3$: 4^n

Series containing Binomial coeff.

Type-0 : coeff exponential

Ex. $\displaystyle\sum_{r=0}^{6} 3^r\,{}^{6}C_r = (1+3)^6 =$ (4096)

Type-I : coeff in numerator

(i) $\sum r\,{}^{n}C_r = n\sum {}^{n-1}C_{r-1} = n\cdot 2^{n-1}$

(ii) $\sum r^2\,{}^{n}C_r$: $r^2 = r(r-1) + r \Rightarrow n(n+1)2^{n-2}$

☆ (iii) $\sum(-1)^r\,{}^{n}C_r\,r^m = 0$ and $\sum(a+rd)^m(-1)^r\,{}^{n}C_r = 0$ $(m < n)$

Ex. $\sum_{r=0}^{9} r\,{}^{9}C_r = 9\cdot 2^8 =$ (2304) , $\sum_{r=0}^{6}(-1)^r\,{}^{6}C_r(r^3+1) =$ (0)

Type-2 : coeff in denominator

(i) $\dfrac{{}^nC_r}{r+1} = \dfrac{{}^{n+1}C_{r+1}}{n+1} \Rightarrow \sum \dfrac{{}^nC_r}{r+1} = \dfrac{2^{n+1}-1}{n+1}$

(ii) $\sum \dfrac{(-1)^r\,{}^nC_r}{r+1} = \dfrac{1}{n+1}$; $\sum \dfrac{(-1)^r\,{}^nC_r}{r+2} = \dfrac{1}{(n+1)(n+2)}$

Doosra tarika : $(1+x)^n$ ko integrate karo (0 to 1)

Ex. $n=7$: $\sum \dfrac{(-1)^r\,{}^7C_r}{r+1} = \boxed{\dfrac{1}{8}}$

Type-3 : product of coefficients

☆ (i) $\sum_r {}^mC_r\,{}^nC_{k-r} = {}^{m+n}C_k$ (Vandermonde)

(ii) $\sum({}^nC_r)^2 = {}^{2n}C_n$; $\sum {}^nC_r\,{}^nC_{r+k} = {}^{2n}C_{n+k}$

(iii) $\sum(-1)^r({}^nC_r)^2 = 0$ (n odd), $(-1)^m\,{}^{2m}C_m$ ($n=2m$)

Ex. $\sum_{r=0}^{7} {}^7C_r\,{}^7C_{r+1} = {}^{14}C_8 = \boxed{3003}$

Roots of unity filter

(a) $1+\omega+\omega^2 = 0$, $\omega^3 = 1$

(b) $(1+1)^n + (1+\omega)^n + (1+\omega^2)^n = 3\,[C_0 + C_3 + C_6 + \dots]$

(c) $(1+1)^n + (1+i)^n + (1-1)^n + (1-i)^n = 4\,[C_0 + C_4 + C_8 + \dots]$

(i) $\sum_k {}^nC_{3k+j} = \dfrac{2^n + 2\cos\frac{(n-2j)\pi}{3}}{3}$ $(j = 0, 1, 2)$

(ii) $\sum_k {}^nC_{4k} = \dfrac{2^n + 2\cdot 2^{n/2}\cos\frac{n\pi}{4}}{4}$ $(n \geq 1)$

Ex. $\sum_k {}^{10}C_{3k} = \dfrac{1024 + 2\cos\frac{10\pi}{3}}{3} = \boxed{341}$

Trinomial & Multinomial

(i) $(1+x+x^2)^n$: $2n+1$ terms, coeffs symmetric, sum $= 3^n$

(ii) even-place sum $= \dfrac{3^n+1}{2}$, odd-place sum $= \dfrac{3^n-1}{2}$

(iii) $(x_1+\cdots+x_k)^n = \sum \dfrac{n!}{n_1!\,n_2!\cdots n_k!} x_1^{n_1}\cdots x_k^{n_k}$

(iv) total terms $= {}^{n+k-1}C_{k-1}$

Ex. Coeff of x^2y^3z in $(x+y+z)^6 = \dfrac{6!}{2!\,3!\,1!} = \boxed{60}$

More identities

(i) $\sum r(r-1)\,{}^nC_r = n(n-1)2^{n-2}$; $\sum (r+1)\,{}^nC_r = (n+2)2^{n-1}$

(ii) $\sum r\,({}^nC_r)^2 = n\,{}^{2n-1}C_{n-1}$; $\sum {}^nC_r\,{}^mC_r = {}^{m+n}C_n$

(iii) Proof of (ii) idea : $(1+x)^m(1+x)^n = (1+x)^{m+n}$, compare x^k

(iv) Alternating squares : $(1-x)^n(1+x)^n = (1-x^2)^n$, compare x^n

(v) $\sum_{r=0}^{n}(-1)^r\,{}^nC_r = 0$ so $\sum_{r=0}^{k}(-1)^r\,{}^nC_r = (-1)^k\,{}^{n-1}C_k$

Ex. $\sum_{r=0}^{6}(r+1)\,{}^6C_r = 8\cdot 2^5 = \boxed{256}$

Number of terms

(i) $(x+a+b)^n$: $\dfrac{(n+1)(n+2)}{2}$ terms ; $(a+b+c+d)^n$: ${}^{n+3}C_3$

(ii) $(a+b+c)^n$ ko $\big((a+b)+c\big)^n$ banakar do baar binomial bhi laga sakte hain

Ex. $(x+y+z)^8$: $\dfrac{9\cdot 10}{2} = {}^{10}C_2 = \boxed{45}$

Examples on sums

Ex. $\sum_{r=0}^{8}(r+1)\,{}^{8}C_r = (8+2)\cdot 2^7 =$ $\boxed{1280}$

Ex. $\sum r(r-1)\,{}^{n}C_r = n(n-1)2^{n-2}$; $\sum_{r=0}^{4}{}^{10}C_r = \dfrac{2^{10} - {}^{10}C_5}{2} =$ $\boxed{386}$

Ex. $\sum_{r=0}^{5} r\,({}^{5}C_r)^2 = 5\cdot {}^{9}C_4 =$ $\boxed{630}$

Ex. $\sum_k {}^{7}C_{3k} = 1+35+7 = \dfrac{2^7 + 2\cos\frac{7\pi}{3}}{3} =$ $\boxed{43}$

Ex. $\sum_k {}^{8}C_{3k+1} = \dfrac{256 + 2\cos\frac{6\pi}{3}}{3} =$ $\boxed{86}$

Ex. $\sum_k {}^{10}C_{4k} = \dfrac{1024 + 2\cdot 32\cos\frac{5\pi}{2}}{4} =$ $\boxed{256}$

Ex. $(1+x+x^2)^4$ *coeffs* : $1, 4, 10, 16, 19, 16, 10, 4, 1$ *(symmetric)*

Complex numbers + Binomial

(i) $(1+\cos\theta + i\sin\theta)^n = \left(2\cos\frac{\theta}{2}\right)^n \left(\cos\frac{n\theta}{2} + i\sin\frac{n\theta}{2}\right)$

(ii) $\sum {}^{n}C_r \cos r\theta = 2^n \cos^n\frac{\theta}{2}\cos\frac{n\theta}{2}$

(iii) $\sum {}^{n}C_r \sin r\theta = 2^n \cos^n\frac{\theta}{2}\sin\frac{n\theta}{2}$

Remainder / Divisibility : $(M\pm 1)^n$ *banao*

(i) $a^n - b^n$ *divisible by* $(a-b)$; $a^n + b^n$ *(n odd) by* $(a+b)$

(ii) *Fermat :* p *prime,* $p \nmid a \Rightarrow a^{p-1} \equiv 1 \pmod p$; *Euler :* $a^{\phi(m)} \equiv 1$

(iii) $(1+x)^n$ *me* x *divisor ka multiple ho to* $x^2, x^3, \ldots$ *wale terms bhi multiple*

Ex. $4^{2022} = (15+1)^{1011} \Rightarrow$ *remainder mod* $15 =$ $\boxed{1}$

Ex. *Last two digits of* $7^{100} = (50-1)^{50} =$ $\boxed{01}$

Ex. $19^{200} + 23^{200}$ *mod* 49 : $(21\mp 2)^{200} \Rightarrow 2^{201} \equiv$ $\boxed{29}$

Ex. $6^n - 5n - 1 = (1+5)^n - 5n - 1$ *divisible by* $\boxed{25}$

More on remainders

Ex. $3^{2n+2} - 8n - 9 = (1+8)^{n+1} - 8n - 9$ divisible by 64

Ex. $2^{3n} - 7n - 1 = (1+7)^n - 7n - 1$ divisible by 49

Ex. 10^{100} mod 11 : $(11-1)^{100} \equiv 1 \Rightarrow 1$

Ex. 5^{99} mod 13 : $5^4 \equiv 1 \Rightarrow 5^{99} \equiv 5^3 = 8$

Ex. 2023^{2023} mod 35 : $2023 \equiv -7$, cycle of 7^k has length $4 \Rightarrow 7$

Remainder hamesha $0 \le R < d$ me do (negative aaye to divisor jod do)

Integral & fractional part : $(a+\sqrt{b})^n = I + f$

(0) $[x] + \{x\} = x$, $\{x\} \in [0,1)$; $I = [R]$, $f = \{R\}$

(i) $g = (a-\sqrt{b})^n$; $(a+\sqrt{b})^n + (a-\sqrt{b})^n$ is an integer

(ii) $0 < a - \sqrt{b} < 1 \Rightarrow f + g = 1$, $(I+f)(1-f) = (a^2-b)^n$

(iii) $-1 < a - \sqrt{b} < 0$, n odd $\Rightarrow f = -g$ sign dekho

Ex. $(7+4\sqrt{3})^6 = I + f$: $(I+f)(1-f) = 1$

Ex. $(\sqrt{2}+1)^5 = I + f$: $I = 82$, $f = (\sqrt{2}-1)^5$

Parity of I

(i) $S_n = (a+\sqrt{b})^n + (a-\sqrt{b})^n$ even and $f + g = 1 \Rightarrow I = S_n - 1$ odd

Ex. $(5+2\sqrt{6})^n$: $S_n = 10S_{n-1} - S_{n-2}$, $S_0 = 2, S_1 = 10$ (all even) $\Rightarrow I$ odd

Agar $a - \sqrt{b} < 0$, n odd : $R - G$ integer hota hai, $R + G$ nahi

Negative / fractional index : $|x| < 1$

(i) $(1+x)^n = 1 + nx + \dfrac{n(n-1)}{2!}x^2 + \dfrac{n(n-1)(n-2)}{3!}x^3 + \dots$

(ii) $(1-x)^{-1} = 1 + x + x^2 + \dots$; $(1-x)^{-2} = \sum(r+1)x^r$

(iii) $(1-x)^{-n} = \sum {}^{n+r-1}C_r\, x^r$; $(1-4x)^{-1/2} = \sum {}^{2r}C_r\, x^r$

(iv) approx : $(1+x)^n \approx 1 + nx$ ($|x|$ very small)

(v) $e^x = \sum \dfrac{x^r}{r!}$; $\ln(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \dots$

Ex. Coeff of x^3 in $(1-4x)^{-1/2} = {}^6C_3 =$ $\boxed{20}$, $\sum \dfrac{r+1}{3^r} =$ $\boxed{\dfrac{9}{4}}$

Expansions : more

(i) $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$; $(1-x)^{-3} = \sum {}^{r+2}C_2\, x^r$

(ii) Pehle bracket ko $(1+x)$ form me lao : $(2+x)^{-1} = \dfrac{1}{2}\left(1 + \dfrac{x}{2}\right)^{-1}$

(iii) Finite hai tabhi jab $n \in \mathbb{N}$; warna infinite terms, $|x| < 1$ zaroori

(iv) $(1+x)^n \approx 1 + nx$; zyada accuracy ke liye $\frac{n(n-1)}{2}x^2$ bhi lo

Mix & match

(i) $(1+x)^n$ me coeff of x^r = coeff of x^{r+1} $\iff$ $r = \frac{n-1}{2}$ (n odd)

(ii) $(x+y)^n$ ke har term me powers ka sum $= n$

Ex. $(1+x)^{20}$: coeffs of x^k, x^{k+1} in ratio $1:2$ $\Rightarrow$ $\frac{20-k}{k+1} = 2$ $\Rightarrow k =$ $\boxed{6}$

Ex. Check : ${}^{20}C_6 = 38760$, ${}^{20}C_7 = 77520$

Examples on expansions

Ex. $(0.98)^5 = (1 - 0.02)^5 \approx 1 - 0.10 =$ $\boxed{0.90}$ (exact 0.9039)

Ex. $(1.25)^{-1/2} = 1 - \frac{1}{8} + \frac{3}{128} - \frac{5}{1024} + \dots \approx$ $\boxed{0.894}$

Ex. $(1-x)^{-3} = \sum \dfrac{(r+1)(r+2)}{2}x^r = \sum {}^{r+2}C_2\, x^r$

Ex. $\ln(1+x)$ ya e^x : denominator me $r!$ ho to e^x, sirf r ho to $\ln$

Good Ques. : JEE Main style

Q1. $\left(x-\frac{1}{x}\right)^{10}$ me term independent of x ? — Ans : -252

Q2. $\sum_{r=0}^{9}\frac{(-1)^r\,{}^9C_r}{r+1}=\,?$ — Ans : $\frac{1}{10}$

Q3. $6^{83}+8^{83}$ ko 49 se divide karne par remainder ? — Ans : 35

Q4. $(\sqrt[3]{2}+\sqrt{3})^{50}$ me kitne terms rational hain ? — Ans : 9

Q5. $(2+3x)^{10}$ ka numerically greatest term at $x=1$? — Ans : $T_7=2449440$

Q6. $\sum_{r=0}^{10}({}^{10}C_r)^2=\,?$ — Ans : 184756

Q7. $(1+x)^{21}+\cdots+(1+x)^{30}$ me x^5 ka coeff ? — Ans : 682017

Q8. $(1+x)^n$ ke $5^{th},6^{th},7^{th}$ coeff A.P. me ho to n ke values ka sum ? — Ans : 21

Good Ques. (Advanced) : JEE Adv style

Q1. $\sum_{r=0}^{8}r^2\,{}^8C_r=\,?$ — Ans : 4608

Q2. $\sum_{r=0}^{5}(-1)^r\,{}^5C_r(2r+3)^4=\,?$ — Ans : 0

Q3. $\sum_{r=0}^{5}{}^8C_r\,{}^8C_{r+3}=\,?$ — Ans : 4368

Q4. $\sum_{r=0}^{6}(-1)^r({}^6C_r)^2=\,?$ — Ans : -20

Q5. $(3+\sqrt{7})^6=I+f$ ho to $(I+f)(1-f)=\,?$ — Ans : 64

Q6. $(1+x+x^2)^7$ me even-power coeffs ka sum ? — Ans : 1094

Kaunsa method lagaun ?

Particular term / coeff / independent of x	$\Rightarrow T_{r+1}$ likho, power barabar karo
Greatest term	$\Rightarrow m = \frac{n+1}{1+\lvert a/b\rvert}$
r in numerator	$\Rightarrow r\,{}^nC_r = n\,{}^{n-1}C_{r-1}$
$(r+1)$ in denominator	$\Rightarrow \frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$ ya integrate
$\sum(-1)^r {}^nC_r P(r)$	$\Rightarrow \deg P < n$ ho to 0
Product of coeffs	$\Rightarrow$ Vandermonde
Every 2nd / 3rd / 4th term	$\Rightarrow x = \pm 1\ ;\ \omega\ ;\ i$
Remainder / last digits	$\Rightarrow (M \pm 1)^n$ ya Fermat
$(a+\sqrt{b})^n = I + f$	$\Rightarrow$ conjugate lo, $f + g = 1$
n negative / fraction	$\Rightarrow \lvert x\rvert < 1$, infinite series

Last minute formula strip

$T_{r+1} = {}^nC_r x^{n-r} y^r$	$\sum {}^nC_r = 2^n$	$\sum r\,{}^nC_r = n2^{n-1}$
$\sum({}^nC_r)^2 = {}^{2n}C_n$	$\frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$	$(1-x)^{-1} = \sum x^r$

Dhyan rakho : common galtiyan

(1) T_{r+1} me r se shuru hota hai, T_r se nahi

(2) $(x-a)^n$ me $(-a)^r$ aata hai

(3) Middle term : n even / odd alag

(4) Greatest term me $[m]+1$ lo, $[m]$ nahi

(5) Remainder negative aaye to divisor jod do

(6) Negative index me $\lvert x\rvert < 1$ check karo

Hints (separately) : JEE Main

Q1 $T_{r+1} = {}^{10}C_r\, x^{10-r}(-1/x)^r$, power $10 - 2r = 0$.

Q2 $\frac{{}^{9}C_r}{r+1} = \frac{{}^{10}C_{r+1}}{10}$, phir $\sum(-1)^k {}^{10}C_k = 0$.

Q3 $6 = 7 - 1,\ 8 = 7 + 1$ likho, 7^2 wale terms hata do.

Q4 $\frac{50-r}{3}$ aur $\frac{r}{2}$ dono integer, $0 \le r \le 50$.

Q5 $m = \frac{n+1}{1+|a/b|}$, integer hai ya nahi dekho.

Q6 Vandermonde ka special case.

Q7 Geometric series $\frac{(1+x)^{31}-(1+x)^{21}}{x}$, x^5 coeff.

Q8 A.P. ka condition $n^2 - (4r+1)n + 4r^2 - 2 = 0$, $r = 5$.

Hints (Advanced) : JEE Adv

Q1 $r^2 = r(r-1) + r$ todo.

Q2 Degree 5 se chhota hai $\rightarrow \sum(-1)^r {}^{n}C_r P(r) = 0$.

Q3 $\sum {}^{n}C_r\, {}^{n}C_{r+k} = {}^{2n}C_{n+k}$.

Q4 $n = 2m$ even, $(-1)^m {}^{2m}C_m$.

Q5 $3 - \sqrt{7} \in (0,1)$, $(I+f)g = (a^2 - b)^n$.

Q6 $\frac{f(1)+f(-1)}{2}$ lagao.

Advanced me kya dhyan rakhein

(i) Identity prove karni ho to $(1+x)^n$ ke coefficient compare karo

(ii) ω / i filter me $\cos$ ka argument sahi likho

(iii) Integral-fractional part me $a - \sqrt{b}$ ka sign dekho

(iv) Sum ko chhote n par check karo